

SPECIFICATIONS

Electrical characteristics at VDD_GPO = 3.3 V, VDD_INTERFACE = 1.8 V, and all other VDDx pins = 1.3 V, T_A = 25°C, unless otherwise noted.

Table 1.

Parameter ¹	Symbol	Min	Typ	Max	Unit	Test Conditions/ Comments
RECEIVERS, GENERAL						
Center Frequency		70		6000	MHz	
Gain						
Minimum			0		dB	
Maximum			74.5		dB	At 800 MHz
			73.0		dB	At 2300 MHz (RX1A, RX2A)
			72.0		dB	At 2300 MHz (RX1B, RX1C, RX2B, RX2C)
			65.5		dB	At 5500 MHz (RX1A, RX2A)
Gain Step			1		dB	
Received Signal Strength Indicator	RSSI					
Range			100		dB	
Accuracy			±2		dB	
RECEIVERS, 800 MHz						
Noise Figure	NF		2		dB	Maximum RX gain
Third-Order Input Intermodulation Intercept Point	IIP3		-18		dBm	Maximum RX gain
Second-Order Input Intermodulation Intercept Point	IIP2		40		dBm	Maximum RX gain
Local Oscillator (LO) Leakage			-122		dBm	At RX front-end input
Quadrature						
Gain Error			0.2		%	
Phase Error			0.2		Degrees	
Modulation Accuracy (EVM)			-42		dB	19.2 MHz reference clock
Input S ₁₁			-10		dB	
RX1 to RX2 Isolation						
RX1A to RX2A, RX1C to RX2C			70		dB	
RX1B to RX2B			55		dB	
RX2 to RX1 Isolation						
RX2A to RX1A, RX2C to RX1C			70		dB	
RX2B to RX1B			55		dB	
RECEIVERS, 2.4 GHz						
Noise Figure	NF		3		dB	Maximum RX gain
Third-Order Input Intermodulation Intercept Point	IIP3		-14		dBm	Maximum RX gain
Second-Order Input Intermodulation Intercept Point	IIP2		45		dBm	Maximum RX gain
LO Leakage			-110		dBm	At receiver front-end input
Quadrature						
Gain Error			0.2		%	
Phase Error			0.2		Degrees	
Modulation Accuracy (EVM)			-42		dB	40 MHz reference clock
Input S ₁₁			-10		dB	
RX1 to RX2 Isolation						
RX1A to RX2A, RX1C to RX2C			65		dB	

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Table 1. (Continued)

Parameter ¹	Symbol	Min	Typ	Max	Unit	Test Conditions/ Comments
RX1B to RX2B			50		dB	
RX2 to RX1 Isolation						
RX2A to RX1A, RX2C to RX1C			65		dB	
RX2B to RX1B			50		dB	
RECEIVERS, 5.5 GHz						
Noise Figure	NF		3.8		dB	Maximum RX gain
Third-Order Input Intermodulation Intercept Point	IIP3		-17		dBm	Maximum RX gain
Second-Order Input Intermodulation Intercept Point	IIP2		42		dBm	Maximum RX gain
LO Leakage			-95		dBm	At RX front-end input
Quadrature						
Gain Error			0.2		%	
Phase Error			0.2		Degrees	
Modulation Accuracy (EVM)			-37		dB	40 MHz reference clock (doubled internally for RF synthesizer)
Input S ₁₁			-10		dB	
RX1A to RX2A Isolation			52		dB	
RX2A to RX1A Isolation			52		dB	
TRANSMITTERS—GENERAL						
Center Frequency		46.875		6000	MHz	
Power Control Range			90		dB	
Power Control Resolution			0.25		dB	
TRANSMITTERS, 800 MHz						
Output S ₂₂			-10		dB	
Maximum Output Power			8		dBm	1 MHz tone into 50 Ω load
Modulation Accuracy (EVM)			-40		dB	19.2 MHz reference clock
Third-Order Output Intermodulation Intercept Point	OIP3		23		dBm	
Carrier Leakage			-50		dBc	0 dB attenuation
			-32		dBc	40 dB attenuation
Noise Floor			-157		dBm/Hz	90 MHz offset
Isolation						
TX1 to TX2			50		dB	
TX2 to TX1			50		dB	
TRANSMITTERS, 2.4 GHz						
Output S ₂₂			-10		dB	
Maximum Output Power			7.5		dBm	1 MHz tone into 50 Ω load
Modulation Accuracy (EVM)			-40		dB	40 MHz reference clock
Third-Order Output Intermodulation Intercept Point	OIP3		19		dBm	
Carrier Leakage			-50		dBc	0 dB attenuation
			-32		dBc	40 dB attenuation
Noise Floor			-156		dBm/Hz	90 MHz offset
Isolation						
TX1 to TX2			50		dB	

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Table 1. (Continued)

Parameter ¹	Symbol	Min	Typ	Max	Unit	Test Conditions/ Comments
TX2 to TX1			50		dB	
TRANSMITTERS, 5.5 GHz						
Output S_{22}			-10		dB	
Maximum Output Power			6.5		dBm	7 MHz tone into 50 Ω load
Modulation Accuracy (EVM)			-36		dB	40 MHz reference clock (doubled internally for RF synthesizer)
Third-Order Output Intermodulation Intercept Point	OIP3		17		dBm	
Carrier Leakage			-50		dBc	0 dB attenuation
			-30		dBc	40 dB attenuation
Noise Floor			-151.5		dBm/Hz	90 MHz offset
Isolation						
TX1 to TX2			50		dB	
TX2 to TX1			50		dB	
TX MONITOR INPUTS (TX_MON1, TX_MON2)						
Maximum Input Level			4		dBm	
Dynamic Range			66		dB	
Accuracy			1		dB	
LO SYNTHESIZER						
LO Frequency Step			2.4		Hz	2.4 GHz, 40 MHz reference clock
Integrated Phase Noise						
800 MHz			0.13		$^{\circ}$ rms	100 Hz to 100 MHz, 30.72 MHz reference clock (doubled internally for RF synthesizer)
2.4 GHz			0.37		$^{\circ}$ rms	100 Hz to 100 MHz, 40 MHz reference clock
5.5 GHz			0.59		$^{\circ}$ rms	100 Hz to 100 MHz, 40 MHz reference clock (doubled internally for RF synthesizer)
REFERENCE CLOCK (REF_CLK)						
Input						REF_CLK is either the input to the XTALP/XTALN pins or a line directly to the XTALN pin
Frequency Range						
Crystal		19		50	MHz	
External Oscillator		20		80	MHz	
Signal Level		0.8		1.3	V p-p	AC-coupled external oscillator. Larger swings close to max level give best performance.
AUXILIARY CONVERTERS						
ADC						
Resolution			12		Bits	

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Table 1. (Continued)

Parameter ¹	Symbol	Min	Typ	Max	Unit	Test Conditions/ Comments
Input Voltage						
Minimum			0.05		V	
Maximum			VDDA1P3_BB - 0.05		V	
DAC						
Resolution			10		Bits	
Output Voltage						
Minimum			0.5		V	
Maximum			VDD_GPO - 0.3		V	
Output Current			10		mA	
DIGITAL SPECIFICATIONS (CMOS)						
Logic Inputs						
Input Voltage						
High		VDD_INTERFACE × 0.8		VDD_INTERFACE	V	
Low		0		VDD_INTERFACE × 0.2	V	
Input Current						
High		-10		+10	μA	
Low		-10		+10	μA	
Logic Outputs						
Output Voltage						
High		VDD_INTERFACE × 0.8			V	
Low				VDD_INTERFACE × 0.2	V	
DIGITAL SPECIFICATIONS (LVDS)						
Logic Inputs						
Input Voltage Range		825		1575	mV	Each differential input in the pair
Input Differential Voltage Threshold		-100		+100	mV	
Receiver Differential Input Impedance			100		Ω	
Logic Outputs						
Output Voltage						
High				1375	mV	
Low		1025			mV	
Output Differential Voltage		150			mV	Programmable in 75 mV steps
Output Offset Voltage			1200		mV	
GENERAL-PURPOSE OUTPUTS						
Output Voltage						
High		VDD_GPO × 0.8			V	
Low				VDD_GPO × 0.2	V	
Output Current			10		mA	
SPI TIMING						VDD_INTERFACE = 1.8 V
SPI_CLK						
Period	t _{CP}	20			ns	
Pulse Width	t _{MP}	9			ns	
SPI_ENB Setup to First SPI_CLK Rising Edge	t _{SC}	1			ns	

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Table 1. (Continued)

Parameter ¹	Symbol	Min	Typ	Max	Unit	Test Conditions/ Comments
Last SPI_CLK Falling Edge to SPI_ENB Hold	t_{HC}	0			ns	
SPI_DI						
Data Input Setup to SPI_CLK	t_S	2			ns	
Data Input Hold to SPI_CLK	t_H	1			ns	
SPI_CLK Rising Edge to Output Data Delay						
4-Wire Mode	t_{CO}	3		8	ns	
3-Wire Mode	t_{CO}	3		8	ns	
Bus Turnaround Time, Read	t_{HZM}	t_H		$t_{CO(max)}$	ns	After BBP drives the last address bit
Bus Turnaround Time, Read	t_{HZS}	0		$t_{CO(max)}$	ns	After AD9361 drives the last data bit
DIGITAL DATA TIMING (CMOS), VDD_INTERFACE = 1.8 V						
DATA_CLK Clock Period	t_{CP}	16.276			ns	61.44 MHz
DATA_CLK and FB_CLK Pulse Width TX Data	t_{MP}	45% of t_{CP}		55% of t_{CP}	ns	
Setup to FB_CLK	t_{STX}	1			ns	TX_FRAME, P0_D, and P1_D
Hold to FB_CLK	t_{HTX}	0			ns	
DATA_CLK to Data Bus Output Delay	t_{DDR}	0		1.5	ns	
DATA_CLK to RX_FRAME Delay Pulse Width	t_{DDV}	0		1.0	ns	
ENABLE	t_{ENPW}	t_{CP}			ns	
TXNRX	$t_{TXNRXPW}$	t_{CP}			ns	FDD independent ENSM mode
TXNRX Setup to ENABLE	$t_{TXNRXSU}$	0			ns	TDD ENSM mode
Bus Turnaround Time Before RX	t_{RPRE}	$2 \times t_{CP}$			ns	TDD mode
After RX	t_{RPST}	$2 \times t_{CP}$			ns	TDD mode
Capacitive Load			3		pF	
Capacitive Input			3		pF	
DIGITAL DATA TIMING (CMOS), VDD_INTERFACE = 2.5 V						
DATA_CLK Clock Period	t_{CP}	16.276			ns	61.44 MHz
DATA_CLK and FB_CLK Pulse Width TX Data	t_{MP}	45% of t_{CP}		55% of t_{CP}	ns	
Setup to FB_CLK	t_{STX}	1			ns	TX_FRAME, P0_D, and P1_D
Hold to FB_CLK	t_{HTX}	0			ns	
DATA_CLK to Data Bus Output Delay	t_{DDR}	0		1.2	ns	
DATA_CLK to RX_FRAME Delay Pulse Width	t_{DDV}	0		1.0	ns	
ENABLE	t_{ENPW}	t_{CP}			ns	
TXNRX	$t_{TXNRXPW}$	t_{CP}			ns	FDD independent ENSM mode
TXNRX Setup to ENABLE	$t_{TXNRXSU}$	0			ns	TDD ENSM mode

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Table 1. (Continued)

Parameter ¹	Symbol	Min	Typ	Max	Unit	Test Conditions/ Comments
Bus Turnaround Time						
Before RX	t_{RPRE}	$2 \times t_{CP}$			ns	TDD mode
After RX	t_{RPST}	$2 \times t_{CP}$			ns	TDD mode
Capacitive Load			3		pF	
Capacitive Input			3		pF	
DIGITAL DATA TIMING (LVDS)						
DATA_CLK Clock Period	t_{CP}	4.069			ns	245.76 MHz
DATA_CLK and FB_CLK Pulse Width	t_{MP}	45% of t_{CP}		55% of t_{CP}	ns	
TX Data						TX_FRAME and TX_D
Setup to FB_CLK	t_{STX}	1			ns	
Hold to FB_CLK	t_{HTX}	0			ns	
DATA_CLK to Data Bus Output Delay	t_{DDR}	0.25		1.25	ns	
DATA_CLK to RX_FRAME Delay	t_{DDV}	0.25		1.25	ns	
Pulse Width						
ENABLE	t_{ENPW}	t_{CP}			ns	
TXNRX	$t_{TXNRXPW}$	t_{CP}			ns	FDD independent ENSM mode
TXNRX Setup to ENABLE	$t_{TXNRXSU}$	0			ns	TDD ENSM mode
Bus Turnaround Time						
Before RX	t_{RPRE}	$2 \times t_{CP}$			ns	
After RX	t_{RPST}	$2 \times t_{CP}$			ns	
Capacitive Load			3		pF	
Capacitive Input			3		pF	
SUPPLY CHARACTERISTICS						
1.3 V Main Supply Voltage		1.267	1.3	1.33	V	
VDD_INTERFACE Supply Nominal Settings						
CMOS		1.14		2.625	V	
LVDS		1.71		2.625	V	
VDD_INTERFACE Tolerance		-5		+5	%	Tolerance is applicable to any voltage setting
VDD_GPO Supply Nominal Setting		1.3		3.3	V	When unused, must be set to 1.3 V
VDD_GPO Tolerance		-5		+5	%	Tolerance is applicable to any voltage setting
Current Consumption						
VDDx, Sleep Mode			180		μA	Sum of all input currents
VDD_GPO			50		μA	No load

¹ When referencing a single function of a multifunction pin in the parameters, only the portion of the pin name that is relevant to the specification is listed. For full pin names of multifunction pins, refer to the [Pin Configuration and Function Descriptions](#) section.